Negotiated Autonomy

Personal AI That Learns How We Delegate Spatial Work

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An action that changes the agreement

A user asks an AI to improve a workspace while keeping the window workstation. The AI also moves a monitor the user expected to remain untouched. It understood the goal, but that understanding did not authorise every means of achieving it.

My research develops personal AI that connects conversation and perceived context with useful action. The question is how people delegate that action: “Move these chairs while I am away” grants latitude; “Do not touch that one” revises it. Such agreements change during work and across sessions.

H1 proposes that at least 80% of positive grants and grant replacements in room-editing tasks fit visible objects or a region, one operation and expiry, under a disclosed default of session end. A fixed coding scheme tests meaning preservation; conditions such as “only if the aisle stays clear” expose its limits.

Connections and the research claim

Reference-Based Manipulation [1] supplies Source, Anchor and Frame for locating permission; operations and duration complete the proposed scope record. EchoLadder [2] makes scene suggestions reversible. This project adds time-limited standing grants and a history of how those grants change across sessions. Orality [3] motivates treating granting, withholding and retracting permission as speech acts within ongoing thought.

Horvitz [4] establishes mixed initiative; automation levels [5] and human-AI guidelines [6] inform execution and correction. The new claim concerns the adequacy of a compact, spatially grounded permission representation and whether negotiating it costs less than editing explicit rules.

What the prototype learns

The desktop/XR prototype links speech and editor state to scene-editing tools. Floating models preview changes; gestures mark their scope. For each user and object-operation class, it learns a default mode: suggest, preview, sandbox or commit. A standing grant lets that default govern repeated actions. An inspectable history records changes, while speech or a palm gesture interrupts execution.

The questions concern how people express delegation, whether dialogue and learned defaults reduce its cost, and how agreements shift over repeated use.

Studies and measures

The first six months will produce a Wizard-of-Oz study with 14 participants and a taxonomy of authorisation speech acts. Naturally expressed grants test H1; targeted counterexamples reveal what the representation leaves out.

Thirty participants will compare three interfaces: an editable rule panel, conversational authorisation, and conversation with learned defaults. Tasks vary how frequently permission changes. I predict that rules cost less when agreements stay stable, while dialogue costs less when they change often. Repeated sessions then compare learning with simply reusing the last approved mode.

Delegation overhead measures the time and interaction needed to authorise work, alongside unwanted actions, completion and repair. Delayed autonomy drift is an action outside the latest explicit grant that remains unnoticed for 120 seconds of active inspection after becoming visible. A six-week deployment with 12 participants tracks how agreements change across sessions.

What I bring

Architectural training and CAVE installation experience ground my spatial interface work. Game Fellows provides a bounded companion prototype and a prepared Solo/Fellow comparison; human evaluation is planned. In my image-detection capstone, I reconstructed 900 decisions from frozen outputs and coefficients, exposing class-specific trade-offs hidden by overall improvement. That experience supports reconstructing an AI action against the permission in force when it occurred.

Selected References

[1] Yangyang He, Zhuangze Hou, Yonglin Chen and Can Liu. 2026. Reference-Based Manipulation: A Framework and Pipeline for Multimodal Spatial Reasoning. UIST 2026 (accepted). doi:10.1145/3830398.3830657

[2] Zhuangze Hou, Jingze Tian, Nianlong Li, Farong Ren and Can Liu. 2025. EchoLadder: Progressive AI-Assisted Design of Immersive VR Scenes. UIST 2025. doi:10.1145/3746059.3747659

[3] Wengxi Li, Jingze Tian and Can Liu. 2026. Orality: A Semantic Canvas for Externalizing and Clarifying Thoughts with Speech. CHI 2026. doi:10.1145/3772318.3791713

[4] Eric Horvitz. 1999. Principles of Mixed-Initiative User Interfaces. CHI 1999, 159-166. doi:10.1145/302979.303030

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[6] Saleema Amershi et al. 2019. Guidelines for Human-AI Interaction. CHI 2019. doi:10.1145/3290605.3300233